Second-Order Linear Homogeneous Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The auxiliary equation**

Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.

$r^2 - 5r + 6 = 0$.

2 **[Foundational]** • **Target: The auxiliary equation**

Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.

$2r^2 + 4r - 6 = 0$, which simplifies to $r^2 + 2r - 3 = 0$.

3 **[Foundational]** • **Target: The auxiliary equation**

Explain why substituting $y = e^{rx}$ into $ay'' + by' + cy = 0$ produces the auxiliary equation $ar^2 + br + c = 0$.

$y = e^{rx}$ gives $y' = re^{rx}$ and $y'' = r^2e^{rx}$. Substituting: $ar^2e^{rx} + bre^{rx} + ce^{rx} = 0$, i.e., $e^{rx}(ar^2 + br + c) = 0$. Since $e^{rx} \neq 0$ for all x, this forces $ar^2 + br + c = 0$.

4 **[Foundational]** • **Target: Distinct real roots**

Solve $y'' - 5y' + 6y = 0$.

$r^2 - 5r + 6 = (r - 2)(r - 3) = 0 \Rightarrow r = 2, 3$. General solution: $y = C_1e^{2x} + C_2e^{3x}$.

5 **[Foundational]** • **Target: Distinct real roots**

Solve $y'' + 2y' - 3y = 0$, $y(0) = 1$, $y'(0) = 1$.

$r^2 + 2r - 3 = (r + 3)(r - 1) = 0 \Rightarrow r = -3, 1$. So $y = C_1e^{-3x} + C_2e^{x}$. From $y(0) = 1$: $C_1 + C_2 = 1$. From $y'(0) = 1$: $-3C_1 + C_2 = 1$. Subtracting gives $4C_1 = 0$, so $C_1 = 0$, $C_2 = 1$. Thus $y = e^x$.

6 **[Foundational]** • **Target: Distinct real roots**

Solve $y'' - y' - 6y = 0$.

$r^2 - r - 6 = (r - 3)(r + 2) = 0 \Rightarrow r = 3, -2$. General solution: $y = C_1e^{3x} + C_2e^{-2x}$.

7 **[Foundational]** • **Target: Distinct real roots**

Solve $y'' - 9y = 0$ (note this factors as a difference of squares in r).

$r^2 - 9 = (r - 3)(r + 3) = 0 \Rightarrow r = \pm 3$. General solution: $y = C_1e^{3x} + C_2e^{-3x}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Repeated roots**

Solve $y'' - 6y' + 9y = 0$.

$r^2 - 6r + 9 = (r-3)^2 = 0 \Rightarrow r = 3$ (repeated). General solution: $y = (C_1 + C_2 x)e^{3x}$.

9 **[Intermediate]** • **Target: Repeated roots**

Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.

$r^2 + 4r + 4 = (r+2)^2 = 0 \Rightarrow r = -2$. So $y = (C_1 + C_2 x)e^{-2x}$. From $y(0) = 2$: $C_1 = 2$. Differentiating, $y' = C_2 e^{-2x} - 2(C_1 + C_2 x)e^{-2x}$; $y'(0) = C_2 - 2C_1 = -1 \Rightarrow C_2 = 2(2) - 1 = 3$. Thus $y = (2 + 3x)e^{-2x}$.

10 **[Intermediate]** • **Target: Repeated roots**

Explain why, for a repeated root r, $y_2 = xe^{rx}$ (not just a second copy of e^{rx}) is needed to build the general solution.

$y_1 = e^{rx}$ and a constant multiple of it, ce^{rx}, are linearly dependent — they don't give two independent solutions needed for a second-order equation's two-dimensional solution space. Direct substitution confirms $y_2 = xe^{rx}$ also solves the ODE when r is a repeated root, and y_1, y_2 are linearly independent (their Wronskian is $e^{2rx} \neq 0$), so together they span the full solution space.

11 **[Intermediate]** • **Target: Complex roots**

Solve $y'' + 4y = 0$.

$r^2 + 4 = 0 \Rightarrow r = \pm 2i$. General solution: $y = C_1 \cos 2x + C_2 \sin 2x$.

12 **[Intermediate]** • **Target: Complex roots**

Solve $y'' - 2y' + 5y = 0$.

$r^2 - 2r + 5 = 0 \Rightarrow r = \frac{2 \pm \sqrt{4-20}}{2} = 1 \pm 2i$. General solution: $y = e^x(C_1 \cos 2x + C_2 \sin 2x)$.

13 **[Intermediate]** • **Target: Complex roots**

Solve $y'' + 9y = 0$, $y(0) = 2$, $y'(0) = 3$.

$r^2 + 9 = 0 \Rightarrow r = \pm 3i$. So $y = C_1 \cos 3x + C_2 \sin 3x$. $y(0) = C_1 = 2$. $y' = -3C_1 \sin 3x + 3C_2 \cos 3x$; $y'(0) = 3C_2 = 3 \Rightarrow C_2 = 1$. Thus $y = 2\cos 3x + \sin 3x$.

14 **[Intermediate]** • **Target: Complex roots**

Solve $y'' + 2y' + 2y = 0$.

$r^2 + 2r + 2 = 0 \Rightarrow r = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$. General solution: $y = e^{-x}(C_1 \cos x + C_2 \sin x)$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 1.01 Separable Equations

Without solving, state whether $y'' + 6y' + 5y = 0$ has real, repeated, or complex roots, using the discriminant $b^2 - 4ac$ of the auxiliary equation.

Discriminant $= 36 - 20 = 16 > 0$ and a perfect square, so the roots are real and distinct ($r = -1, -5$).

16 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Without solving, classify the roots of $y'' + 10y' + 25y = 0$.

Discriminant $= 100 - 100 = 0$, so the auxiliary equation has one repeated real root ($r = -5$).

17 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Without solving, classify the roots of $y'' + 2y' + 10y = 0$.

Discriminant $= 4 - 40 = -36 < 0$, so the roots are complex conjugates ($r = -1 \pm 3i$).

18 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 5.01 Variation of Parameters

A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

$r^2 + 4r + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, so $r = -2 \pm i$. General solution: $y = e^{-2x}(C_1 \cos x + C_2 \sin x)$ — this is under-damped (complex roots with negative real part give oscillatory decay).

19 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 1.01 Separable Equations

Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

$r^2 + 6r + 9 = (r + 3)^2 = 0 \Rightarrow r = -3$ (repeated) — critically damped. So $y = (C_1 + C_2 x)e^{-3x}$. From $y(0) = 1$: $C_1 = 1$. $y' = C_2 e^{-3x} - 3(C_1 + C_2 x)e^{-3x}$; $y'(0) = C_2 - 3C_1 = 0 \Rightarrow C_2 = 3$. Thus $y = (1 + 3x)e^{-3x}$.

20 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y'' + 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

$r^2 + 5r + 6 = (r + 2)(r + 3) = 0 \Rightarrow r = -2, -3$ (real, distinct, both negative) — over-damped. So $y = C_1 e^{-2x} + C_2 e^{-3x}$. $y(0) = C_1 + C_2 = 1$. $y' = -2C_1 e^{-2x} - 3C_2 e^{-3x}$; $y'(0) = -2C_1 - 3C_2 = 0 \Rightarrow C_1 = -\frac{3}{2}C_2$. Substituting: $-\frac{3}{2}C_2 + C_2 = 1 \Rightarrow -\frac{1}{2}C_2 = 1 \Rightarrow C_2 = -2, C_1 = 3$. Thus $y = 3e^{-2x} - 2e^{-3x}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distinct Real Roots

Solve the auxiliary characteristic equation for $y'' - 5y' + 6y = 0$.

a. $r = 2$ **and** $r = 3$ ✓

b. $r = -2$ and $r = -3$

c. $r = 1$ and $r = 6$

d. $r = \pm\sqrt{6}$

Characteristic equation: $r^2 - 5r + 6 = 0 \Rightarrow (r-2)(r-3) = 0 \Rightarrow r = 2, 3$.

22 [Foundational] • Target: Distinct Real Roots

What is the general solution to $y'' - 5y' + 6y = 0$?

a. $y = c_1e^{2x} + c_2e^{3x}$ ✓

b. $y = c_1e^{-2x} + c_2e^{-3x}$

c. $y = (c_1 + c_2x)e^{2x}$

d. $y = c_1\cos(2x) + c_2\sin(3x)$

For two distinct real roots r_1, r_2, the fundamental solution set is $\{e^{r_1x}, e^{r_2x}\}$, yielding $y = c_1e^{2x} + c_2e^{3x}$.

23 [Foundational] • Target: Distinct Real Roots

Solve the IVP $y'' - y = 0$ with $y(0) = 4$ and $y'(0) = 2$.

a. $y = 3e^x + e^{-x}$ ✓

b. $y = 2e^x + 2e^{-x}$

c. $y = 4e^x$

d. $y = e^x + 3e^{-x}$

$r^2 - 1 = 0 \Rightarrow r = \pm 1 \Rightarrow y = c_1e^x + c_2e^{-x}$. $y'(x) = c_1e^x - c_2e^{-x}$. $y(0) = c_1 + c_2 = 4$ and $y'(0) = c_1 - c_2 = 2$. Adding: $2c_1 = 6 \Rightarrow c_1 = 3, c_2 = 1$. Thus $y = 3e^x + e^{-x}$.

24 [Foundational] • Target: Repeated Real Roots

What is the general solution to $y'' - 6y' + 9y = 0$?

a. $y = (c_1 + c_2x)e^{3x}$ ✓

b. $y = c_1e^{3x} + c_2e^{-3x}$

c. $y = c_1e^{3x} + c_2x^2e^{3x}$

d. $y = c_1\cos(3x) + c_2\sin(3x)$

Characteristic equation: $r^2 - 6r + 9 = (r-3)^2 = 0 \Rightarrow r = 3$ (repeated root). The general solution is $y = (c_1 + c_2x)e^{3x}$.

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25 [Foundational] • Target: Repeated Real Roots

Why is the factor x introduced into the second solution xe^{rx} for repeated roots?

a. **To ensure linear independence from the first solution e^{rx}** ✓

b. Because the derivative of e^{rx} contains x

c. To make the solution satisfy $y(0) = 0$

d. By Euler's identity

Reduction of order proves that $y_2 = xe^{rx}$ is linearly independent of $y_1 = e^{rx}$, giving a non-zero Wronskian.

26 [Foundational] • Target: Repeated Real Roots

Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = -1$.

a. $y = (2 + 3x)e^{-2x}$ ✓

b. $y = (2 - x)e^{-2x}$

c. $y = 2e^{-2x}$

d. $y = (2 + x)e^{-2x}$

$r = -2$ repeated: $y = (c_1 + c_2x)e^{-2x}$. $y(0) = c_1 = 2$. $y'(x) = c_2e^{-2x} - 2(c_1 + c_2x)e^{-2x}$. $y'(0) = c_2 - 2c_1 = -1 \Rightarrow c_2 - 4 = -1 \Rightarrow c_2 = 3$. Thus $y = (2 + 3x)e^{-2x}$.

27 [Foundational] • Target: Complex Conjugate Roots

If the characteristic equation has complex roots $r = \alpha \pm i\beta$, what is the real general solution?

a. $y = e^{\alpha x}(c_1\cos(\beta x) + c_2\sin(\beta x))$ ✓

b. $y = c_1e^{\alpha x} + c_2e^{\beta x}$

c. $y = e^{\beta x}(c_1\cos(\alpha x) + c_2\sin(\alpha x))$

d. $y = c_1\cos((\alpha + \beta)x)$

By Euler's formula $e^{(\alpha \pm i\beta)x} = e^{\alpha x}(\cos(\beta x) \pm i\sin(\beta x))$, the real fundamental solution basis is $\{e^{\alpha x}\cos(\beta x), e^{\alpha x}\sin(\beta x)\}$.

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28 [Foundational] • Target: Complex Conjugate Roots

Find the general solution to $y'' + 16y = 0$.

a. $y = c_1\cos(4x) + c_2\sin(4x)$ ✓

b. $y = c_1e^{4x} + c_2e^{-4x}$

c. $y = (c_1 + c_2x)e^{4x}$

d. $y = c_1\cos(16x) + c_2\sin(16x)$

$r^2 + 16 = 0 \Rightarrow r = \pm 4i$ (with $\alpha = 0, \beta = 4$). Thus $y = c_1\cos(4x) + c_2\sin(4x)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Complex Conjugate Roots

Solve the characteristic equation for $y'' - 2y' + 5y = 0$.

a. $r = 1 \pm 2i$ ✓

b. $r = 2 \pm 4i$

c. $r = -1 \pm 2i$

d. $r = 1 \pm 4i$

Quadratic formula: $r = \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$.

30 [Intermediate] • Target: Harmonic Motion and Damping Regimes

A mechanical oscillator modeled by $my'' + cy' + ky = 0$ is underdamped when:

a. $c^2 - 4mk < 0$ **(oscillatory motion with exponentially decaying amplitude)** ✓

b. $c^2 - 4mk = 0$ (critically damped)

c. $c^2 - 4mk > 0$ (overdamped, no oscillations)

d. $c = 0$ (undamped simple harmonic motion)

When the discriminant $c^2 - 4mk < 0$, the characteristic roots are complex conjugates, causing oscillatory motion that decays exponentially.

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31 [Intermediate] • Target: Harmonic Motion and Damping Regimes

Which damping condition returns a displaced mass to equilibrium in the shortest time without oscillating?

a. Critical damping ($c^2 = 4mk$) ✓

b. Overdamping ($c^2 > 4mk$)

c. Underdamping

d. Zero damping

Critical damping provides the most rapid return to equilibrium without experiencing overshoot or oscillation.

32 [Intermediate] • Target: Harmonic Motion and Damping Regimes

What is the natural frequency ω_0 of the undamped oscillator $y'' + 25y = 0$?

a. $\omega_0 = 5$ rad/s ✓

b. $\omega_0 = 25$ rad/s

c. $\omega_0 = 10$ rad/s

d. $\omega_0 = 2.5$ rad/s

In $y'' + \omega_0^2 y = 0$, $\omega_0^2 = 25 \Rightarrow \omega_0 = 5$ rad/s.

33 [Intermediate] • Target: The auxiliary equation

Consider the second-order linear homogeneous differential equation: $y''(x) - ky'(x) + 4y(x) = 0$. If the characteristic equation has a repeated real root, and one of the solutions to the differential equation is $y_1(x) = xe^{2x}$, determine the value of the parameter k.

a. 2

b. 4 ✓

c. 3

d. 1

The given differential equation is $y''(x) - ky'(x) + 4y(x) = 0$. The auxiliary equation is $r^2 - kr + 4 = 0$. For the auxiliary equation to have a repeated real root, the discriminant must be zero: $D = (-k)^2 - 4(1)(4) = 0$. This implies $k^2 - 16 = 0$, so $k = \pm 4$. If $k = 4$, the auxiliary equation is $r^2 - 4r + 4 = 0$, which factors as $(r - 2)^2 = 0$, yielding a repeated real root $r = 2$. The general solution for a repeated real root r is $y(x) = c_1 e^{rx} + c_2 xe^{rx}$. In this case.

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34 [Intermediate] • Target: Distinct real roots

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$. \\ If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter k in $y_2(x) = kxe^x$.

a. $k = 5$ ✓

b. $k = -5$

c. $k = 1$

d. $k = -1$

The given differential equation is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r - 1)^2 = 0$. Thus, the roots are repeated: $r_1 = r_2 = 1$. \\ One fundamental solution is given as $y_1(x) = e^x$. For a second-order linear homogeneous equation $y'' + p(x)y' + q(x)y = 0$ with a known solution $y_1(x)$, the Wronskian is given by $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x) = Ce^{-\int p(x)dx}$. \\ In our equation, $p(x) = -2$. So, $-\int p(x)dx = -\int(-2)dx = 2x$..

35 [Intermediate] • Target: Repeated roots

Consider the second-order linear homogeneous differential equation $y'' - 2y' + ky = 0$. We are given that $y_1(x) = e^{3x}$ is a solution to this equation. \\ (a) Find the value of the parameter k. \\ (b) Find the general solution to the differential equation.

a. $k = -3, y(x) = c_1e^{3x} + c_2e^{-x}$ ✓

b. $k = 3, y(x) = c_1e^{3x} + c_2xe^{3x}$

c. $k = -3, y(x) = c_1e^{3x} + c_2e^x$

d. $k = 9, y(x) = c_1e^{3x} + c_2e^{-3x}$

Part (a): \\ If $y_1(x) = e^{3x}$ is a solution, then its derivatives must satisfy the equation.\\ $y_1'(x) = 3e^{3x}$\\ $y_1''(x) = 9e^{3x}$\\ Substitute these into the differential equation $y'' - 2y' + ky = 0$:\\ $9e^{3x} - 2(3e^{3x}) + k(e^{3x}) = 0$\\ $9e^{3x} - 6e^{3x} + ke^{3x} = 0$\\ Factor out e^{3x} (which is never zero):\\ $e^{3x}(9 - 6 + k) = 0$\\ $3 + k = 0$\\ So, $k = -3$.\\ \\ Part (b): \\ Now that we know $k = -3$, the differential equation is $y'' - 2y' - 3y = 0$.\\ The auxiliary (characteristic).

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36 [Intermediate] • Target: Complex roots

Consider the second-order linear homogeneous differential equation $y'' - 3y' + 2y = 0$. Find the general solution to this equation.

a. $y(x) = c_1e^x + c_2e^{2x}$ ✓

b. $y(x) = c_1e^{-x} + c_2e^{-2x}$

c. $y(x) = c_1\cos(x) + c_2\sin(x)$

d. $y(x) = c_1xe^x + c_2e^x$

The characteristic equation for the differential equation $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = 0$. Factoring this quadratic equation gives $(r - 1)(r - 2) = 0$. The roots are $r_1 = 1$ and $r_2 = 2$. Since the roots are real and distinct, the general solution is of the form $y(x) = c_1e^{r_1x} + c_2e^{r_2x}$. Substituting the roots, we get $y(x) = c_1e^x + c_2e^{2x}$. Choice B is incorrect because it uses negative roots. Choice C is incorrect because it corresponds to complex roots. Choice D is incorrect becau.

37 [Intermediate] • Target: Classifying without fully solving

Consider the second-order linear homogeneous differential equation $y'' + P(x)y' + Q(x)y = 0$. For the equation $x^2y'' + xy' + (x^2 - u^2)y = 0$ (Bessel's equation of order u), the point $x = 0$ is a regular singular point. If we assume a series solution of the form $y(x) = \sum_{n=0}^{\infty} a_nx^{n+r}$, where a_0eq0, the indicial equation is given by $r^2 - u^2 = 0$. For this problem, let $u = 1$, so the indicial equation is $r^2 - 1 = 0$. This yields two distinct roots, $r_1 = 1$ and $r_2 = -1$. For the larger root $r_1 = 1$, the power series solution takes the form $y_1(x) = x + \sum_{n=1}^{\infty} a_nx^{n+1}$. Substituting this into Bessel's equation of order 1, and after considerable algebraic manipulation, the recurrence relation for the coefficients a_n is found to be $a_n = -\frac{a_{n-2}}{n(n+2)}$ for $n \geq 2$. Given $a_0 = 1$, find the value of a_3 for the solution corresponding to the larger root $r_1 = 1$.

a. **0** ✓

b. 1/24

c. 1/48

d. 1/12

The recurrence relation is $a_n = -\frac{a_{n-2}}{n(n+2)}$ for $n \geq 2$, with $a_0 = 1$. We need to find a_3. For $n = 2$, $a_2 = -\frac{a_0}{2(2+2)} = -\frac{1}{2(4)} = -\frac{1}{8}$. For $n = 3$, $a_3 = -\frac{a_{3-2}}{3(3+2)} = -\frac{a_1}{3(5)} = -\frac{a_1}{15}$. Since the recurrence relation only relates terms that are two indices apart, and we are given a_0, any odd-indexed coefficients will depend on a_1. However, the form of the solution for Bessel's equation of order u is.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 5.01 Variation of Parameters

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

a. 5 ✓

b. -5

c. 6

d. -6

The given differential equation is $y'' - 5y' + 6y = 0$. The auxiliary equation for a second-order linear homogeneous differential equation of the form $ay'' + by' + cy = 0$ is $ar^2 + br + c = 0$. In this case, $a = 1$, $b = -5$, and $c = 6$. So the auxiliary equation is $r^2 - 5r + 6 = 0$. Let the roots of this quadratic equation be r_1 and r_2. According to Vieta's formulas, for a quadratic equation $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2$ is equal to $-\frac{b}{a}$. In our auxiliary equation.

39 [Challenging] • Target: Verifying general solutions and the superposition principle • Cross-Lesson: 1.01 Separable Equations

Consider the second-order linear homogeneous differential equation $y''(x) + 5y'(x) + 6y(x) = 0$. If the auxiliary equation has roots r_1 and r_2, what is the value of $r_1 + r_2$?

a. 5

b. -5 ✓

c. 6

d. -6

The general form of a second-order linear homogeneous differential equation with constant coefficients is $ay''(x) + by'(x) + cy(x) = 0$. The associated auxiliary equation is $ar^2 + br + c = 0$. For the given equation $y''(x) + 5y'(x) + 6y(x) = 0$, the auxiliary equation is $r^2 + 5r + 6 = 0$. According to Vieta's formulas, for a quadratic equation of the form $ar^2 + br + c = 0$, the sum of the roots $r_1 + r_2 = -b/a$. In this case, $a = 1$, $b = 5$, and $c = 6$. Therefore, the sum of the roots is.

40 [Challenging] • Target: Boundary value problems • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the characteristic equation has roots r_1 and r_2 with $r_1 < r_2$, what is the value of r_1?

a. 1

b. 2 ✓

c. 3

d. 6

The characteristic equation for the given differential equation $y'' - 5y' + 6y = 0$ is found by replacing y'' with r^2, y' with r, and y with 1. This gives us the quadratic equation $r^2 - 5r + 6 = 0$. To find the roots, we can factor this quadratic equation: $(r - 2)(r - 3) = 0$. The roots are therefore $r = 2$ and $r = 3$. We are given that $r_1 < r_2$. Thus, $r_1 = 2$ and $r_2 = 3$. The question asks for the value of r_1, which is 2.

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41 [Challenging] • Target: Distinct Real Roots • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

For which values of k does every nonzero solution of $y'' + 4y' + ky = 0$ genuinely oscillate (the auxiliary equation has non-real roots)?

a. $k < 4$

b. $k = 4$

c. $k > 0$

d. $k > 4$ ✓

The auxiliary equation $r^2 + 4r + k = 0$ has discriminant $16 - 4k$. Non-real roots require $16 - 4k4$ — exactly the range producing Theorem 3.2's complex-root case.

42 [Challenging] • Target: Repeated Real Roots • Cross-Lesson: 5.01 Variation of Parameters

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - u^2)y = 0$, where u is a constant. This is Bessel's equation of order u. Assume $u = 1/2$ and that $x > 0$. x is a regular singular point. Determine the indicial equation for this differential equation and find its roots.

a. **Indicial equation:** $r^2 - \frac{1}{4} = 0$**; roots:** $r = \pm\frac{1}{2}$ ✓

b. Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

c. Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

d. Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

The given differential equation is $x^2y'' + xy' + (x^2 - u^2)y = 0$. We can rewrite this in the standard form $y'' + P(x)y' + Q(x)y = 0$ by dividing by x^2: $y'' + \frac{1}{x}y' + \frac{x^2 - u^2}{x^2}y = 0$. Thus, $P(x) = \frac{1}{x}$ and $Q(x) = \frac{x^2 - u^2}{x^2} = 1 - \frac{u^2}{x^2}$. $x = 0$ is a singular point. To determine if it's a regular singular point, we examine $xP(x)$ and $x^2Q(x)$. $xP(x) = x \cdot \frac{1}{x} = 1$, which is analytic at $x = 0$..

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43 [Challenging] • Target: Complex Conjugate Roots • Cross-Lesson: 1.01 Separable Equations

Consider the second-order linear homogeneous differential equation: $y'' - xy' + y = 0$. The point $x = 0$ is an ordinary point. Find the first two non-zero terms of the power series solution around $x = 0$, assuming $y(0) = 1$ and $y'(0) = 0$.

a. $1 + \frac{1}{2}x^2 + \ldots$ ✓

b. $1 + \frac{1}{3}x^3 + \ldots$

c. $1 - \frac{1}{2}x^2 + \ldots$

d. $1 - \frac{1}{3}x^3 + \ldots$

Let the power series solution be $y(x) = \sum_{n=0}^{\infty} a_n x^n$. Then $y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ and $y''(x) = \sum_{} n(n-1)a_n x^{n-2}$. Substituting into the differential equation $y'' - xy' + y = 0$: $\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - x\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$ $\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$ To combine the series, we shift the index in the first ter.

44 [Challenging] • Target: Harmonic Motion and Damping Regimes • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the nonhomogeneous second-order linear differential equation $y'' - y = \frac{1}{1+e^x}$. If $y_1(x) = e^x$ and $y_2(x) = e^{-x}$ are two linearly independent solutions to the associated homogeneous equation, find the Wronskian of y_1 and y_2.

a. 2

b. **-2** ✓

c. 0

d. $e^x - e^{-x}$

The Wronskian of two functions $y_1(x)$ and $y_2(x)$ is defined as $W(y_1, y_2)(x) = \det(y_1(x), y_2(x)\ y_1'(x), y_2'(x))$. Given $y_1(x) = e^x$ and $y_2(x) = e^{-x}$, their derivatives are $y_1'(x) = e^x$ and $y_2'(x) = -e^{-x}$. Substituting these into the Wronskian formula: $W(y_1, y_2)(x) = \det(e^x, e^{-x}\ e^x,\ -e^{-x})\ W(y_1, y_2)(x) = (e^x)(-e^{-x}) - (e^{-x})(e^x)$ $W(y_1, y_2)(x) = -e^{x-x} - e^{-x+x}\ W(y_1, y_2)(x) = -e^0 - e^0\ W(y_1, y_2)(x) = -1 - 1$.

45 [Challenging] • Target: The auxiliary equation • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A damped spring satisfies $y'' + 2y' + 10y = 0$, $y(0) = 0$, $y'(0) = 6$. Find $y(\pi/6)$.

a. $2e^{\pi/6}$

b. $2e^{-\pi/6}$ ✓

c. $-2e^{-\pi/6}$

d. $e^{-\pi/6}$

Auxiliary: $r^2 + 2r + 10 = 0 \Rightarrow r = \frac{-2 \pm \sqrt{4-40}}{2} = -1 \pm 3i$ — $\alpha = -1, \beta = 3$, so $y = e^{-x}(C_1 \cos 3x + C_2 \sin 3x)$. $y(0) = C_1 = 0$, so $y = C_2 e^{-x} \sin 3x$; $y' = C_2 e^{-x}(3\cos 3x - \sin 3x)$, $y'(0) = 3C_2 = 6 \Rightarrow C_2 = 2$. So $y = 2e^{-x} \sin 3x$, and $y(\pi/6) = 2e^{-\pi/6} \sin(\pi/2) = 2e^{-\pi/6}$.

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